Paper 15.25 - Toolkit for Mass-Residue Carrier

CQE-paper-15.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-15.25.md

Paper 15.25 - Toolkit for Mass-Residue Carrier

Purpose

This support paper exposes the tools for Paper 15. It does not prove the Higgs mechanism. It shows how to inspect the finite substrate residue carrier.

Digital Route

Run:

```
python production/formal-papers/CQE-paper-15/verify_mass_residue_carrier.py
```

The verifier checks:

```
Rule 30 split
F2 Arf invariant
Arf gluing/rejection
VOA weight distribution
correction residue states
nth-bit local/oracle layer with McKay-Thompson parity still open
```

Analog Route

Write the local state as three marks:

```
L C R
```

Cancel the linear part:

```
L xor C xor R
```

The obstruction token is:

```
C and R
```

The correction residue token is:

```
C and not R
```

Color the state by VOA weight:

```
0 = vacuum
5 = excited carrier
```

Hidden-Guess Diagnostic

Before revealing the receipt, ask:

```
Does the state carry correction residue?
Is its VOA weight 0 or 5?
Does its Arf class glue or reject?
Is the claim substrate-level or physical-Higgs-level?
```

Reveal the receipt only after the guess is recorded.

Boundary

This toolkit can prove the local residue carrier. It cannot turn that carrier into a measured particle mass without another proof.